

Karthik Bhattaram

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EDUCATION

University of California, Santa Barbara

Santa Barbara

Master of Science in Computer Science

Jan. 2026 to Mar. 2027

- Courses (planned): Distributed Systems, Advanced Topics in Security, Database Systems

University of California, Santa Barbara

Santa Barbara

Bachelor of Science in Computer Science

Sept. 2022 to Dec. 2025

- Courses: Machine Learning for Networking, Computer Security, Data Structures and Algorithms
- Honors: Regents' Scholar, College of Engineering Honors

EXPERIENCE

Software Engineer Intern

June 2025 – Sept. 2025

Arista Networks

Santa Clara, California

- Developed C++ utilities to parse and interpret hardware registers in Broadcom's switching ASICs
- Created a custom data structure to detect deadlocked Ethernet interfaces and mitigate **PFC pause storms**

Machine Learning Research Intern

Sept. 2024 – June 2025

University of California, Santa Barbara

Santa Barbara, California

- Developed pipelines to automate common network measurement efforts
- Aggregated FCC datasets into a SQL database to enable agentic **retrieval-augmented generation** (RAG)

Software Engineer Intern

July 2024 – Sep. 2024

Cadence Design Systems / OpenEye Scientific

Santa Fe, New Mexico

- Wrote **AWS**-based package to benchmark database compression algorithms
- Updated **CUDA** C++ software using new language features to match competing solutions

Hardware Architecture Intern

June 2023 – Sept. 2023

Cadence Design Systems

San Jose, California

- Developed an automation tool to produce memory wrappers of various dimensions and architectures
- Collected **System Verilog** files to be used as templates for this tool and tested memories it produced

Software Engineer Intern

June 2021 – Aug. 2022

Axiado Corporation

San Jose, California

- Developed a package to create **network packet flows** based on IPv6, ARP, ICMP, etc. in simulation
- Created and executed a test plan for a hardware firewall's parser module and discovered multiple bugs

PROJECTS

College Course Database | *Java, SQL*

March 2025 – May 2025

- Designed a database schema using Entity-Relationship diagrams
- Wrote SQL commands to create the tables and enforce data integrity
- Implemented a Java **backend** with JDBC to manage database transactions
- Created command line interfaces for the registrar and students with distinct permissions

TECHNICAL SKILLS

Certificates: Enterprise Security Professional, Data Science and AI

Networking Technologies: Scapy, Wireshark, TCP/IP, Ethernet

Languages: Python, C/C++, Java, JavaScript, Shell, Verilog

Developer Tools: Git, Visual Studio Code, Vim, Emacs

Libraries: CUDA, Pandas, NumPy, PyTest, Black, Matplotlib